Fine-Tuning a Small LLM (DistilGPT-2 + LoRA) for Customer-Support FAQs

Deep Learning – Module 5 Assignment

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1. Introduction

Generic language models such as DistilGPT-2 know nothing about a particular company's policies. This project fine-tunes a small causal language model on a corpus of customer-support FAQs for an e-commerce store, so that it answers typical customer questions concisely and in line with that store's policies. Only a small amount of compute is used: a model with at most 125 M parameters, LoRA instead of full fine-tuning, mixed precision and a single Colab T4 GPU.

The deliverables of the assignment are covered as follows: cleaned JSONL dataset (Section 2), fine-tuned model with checkpoints and inference script (Sections 3 and 5), manual and automated evaluation (Section 4) and this report with challenges and sample outputs (Section 6).

2. Dataset

Domain and construction. The domain is e-commerce customer support for a fictional store, *ShopEase*. Because no real FAQ corpus was used, the data was synthesised from 56 policy topics (returns, refunds, shipping, payments, accounts, promotions, warranty, contact options and similar). Every topic has six paraphrased questions and two alternative wordings of one consistent answer, and each base question was expanded with polite prefixes and suffixes ("Hi, ...", "... Thanks!") or lower-cased variants. The result is **837 prompt/response pairs** from 336 base questions. All answers follow one policy sheet (for example 30-day returns, refunds in 5–7 business days, free standard shipping over $50, express shipping $12.99). A base model cannot know these store-specific facts, so a correct answer after fine-tuning has to come from the training data.

Format. One JSON object per line, `{"prompt": "How do I track my order?", "response": "You can track your order under My Orders ..."}`, with an additional `qid` field that identifies paraphrases of the same base question.

Cleaning. Unicode normalisation (NFKC), removal of control characters, whitespace collapsing, removal of empty rows, a length filter (questions 2–60 words, answers 3–80 words), removal of rows where question equals answer, and case-insensitive de-duplication of questions. Because the data was generated from clean templates, no row was removed by any step (837 pairs before and after).

Split. 80/10/10 grouped by `qid`, so that paraphrases of one question never appear in both training and test data: **671 training, 85 validation and 81 test pairs**. Test questions therefore have wordings that were not seen in training, although the same policy topics are present in training. With the chosen prompt format the training sequences are short (mean 40 tokens, maximum 58), so nothing was truncated at the 128-token limit.

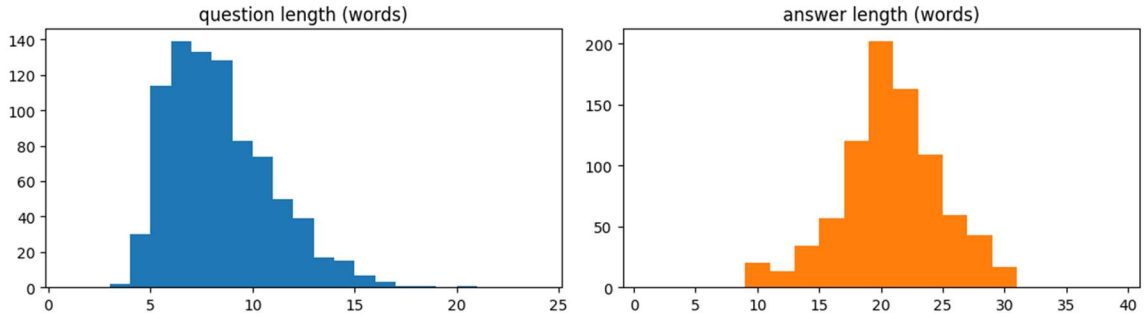


Figure 1 – Distribution of question and answer lengths in words (file: fig_dataset_lengths.png).

3. Model and training setup

Item	Setting
Base model	`distilgpt2`, 81.9 M parameters (limit: 125 M)
Method	LoRA (PEFT): rank 32, alpha 64, dropout 0.05 on the `c_attn`, `c_proj` and `c_fc` layers
Trainable parameters	2,359,296 of 84,271,872 (2.8 %)
Prompt format	`Customer: <question>` newline `Support: <answer><eos>`; the loss is computed on the answer tokens only (question tokens masked)
Trainer	Hugging Face `Trainer`, 5 epochs, batch size 8, learning rate 5e-4 with cosine decay, 20 warm-up steps, weight decay 0.01, fp16 mixed precision
Overfitting control	evaluation and checkpoint every epoch, early stopping (patience 2) on validation loss, best checkpoint restored at the end
Quantisation	not used: bitsandbytes does not quantise the Conv1D layers of GPT-2-style models and the 82 M model fits easily in memory
Hardware and software	Tesla T4, PyTorch 2.11, Transformers 5.20.1, PEFT 0.21.2, seed 42

The minimal-compute strategy of the assignment is therefore met: Colab T4, a model below 125 M parameters, LoRA with mixed-precision training, and a total training time of only **34 seconds** (420 optimisation steps).

Training behaviour. The training loss falls from about 3.7 to about 0.5 and the validation loss from about 2.03 after epoch 1 to 1.31, 1.02, 0.88 and finally **0.864 after epoch 5** (values read from the curves, exact numbers in `training_log.csv`). The best checkpoint was the last one (checkpoint-420), early stopping was never triggered, and the validation curve is clearly flattening: the improvement from epoch 4 to 5 is small. The gap between the training loss (about 0.5) and the validation loss (about 0.86) shows moderate over-fitting to the training wordings. Part of the validation loss is probably irreducible,

because every topic has two equally valid answer wordings and the model cannot know which one a held-out question was paired with.

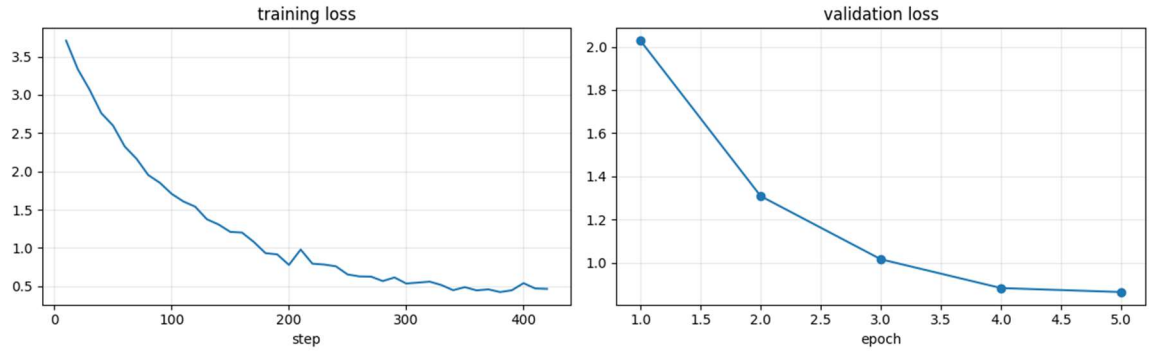


Figure 2 – Training loss per step and validation loss per epoch (file: fig_training_curves.png).

4. Evaluation

4.1 Automated metrics on the test set

The test set has 81 questions. Answers are generated with greedy decoding (maximum 64 new tokens, no repeated 4-grams), identical for both models. The "base model" is the same network with the LoRA adapter switched off, given the same prompt format. Metrics compare the generated answer with the reference answer.

Metric	Base model	Fine-tuned (LoRA)	Change
BLEU (sacreBLEU, 0–100)	0.00	14.70	+14.70
ROUGE-1 (F)	0.007	0.323	+0.316
ROUGE-L (F)	0.006	0.267	+0.262
Embedding cosine similarity (all-MiniLM-L6-v2)	0.067	0.517	+0.450
Test loss (per answer token)	3.717	0.908	−2.808
Perplexity of the reference answers	41.1	2.48	−38.7

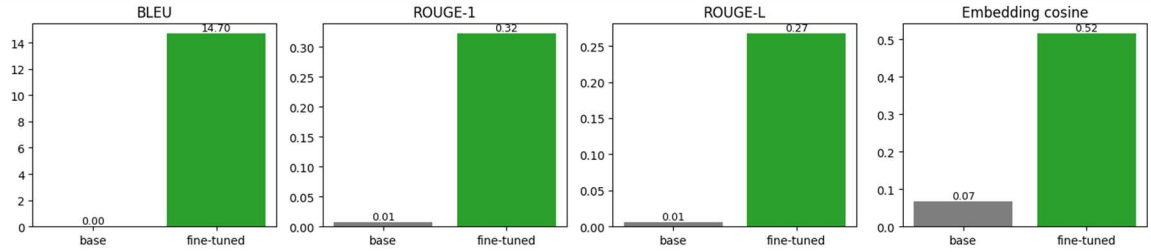


Figure 3 – Base versus fine-tuned model on BLEU, ROUGE-1, ROUGE-L and embedding similarity (file: fig_metrics_base_vs_ft.png).

Reading the numbers. Fine-tuning improves every metric. The most trustworthy comparison is the loss: the perplexity of the correct store answers drops from 41 to 2.5, which means the model has learnt the style and content of the ShopEase answers. The generation-based metrics need two cautions. First, the base model's generations are nearly all empty in this run (the base model starts its continuation with a line break and the notebook keeps only the first line of the generated text), so its BLEU and ROUGE of about zero exaggerate the gap; the loss and perplexity are not affected by this. Second, the absolute scores of the fine-tuned model are moderate (BLEU 14.7, ROUGE-L 0.27, cosine 0.52). BLEU and ROUGE penalise correct answers that are worded differently from the reference, and, as Section 4.2 shows, a large share of the generated answers are genuinely wrong, mostly because the model picks the answer of a neighbouring topic.

4.2 Manual evaluation: base versus fine-tuned on 10 queries

#	Query	Expected (gist)	Base model	Fine-tuned model	Verdict
1	Can I return a product after 30 days?	Returns only within 30 days of delivery	(empty)	"Items must be unused, in their original packaging and with all tags attached to be eligible for a return."	Partly: right topic, does not state the 30-day limit
2	How do I track my order?	My Orders > Track, tracking email	(empty)	"We track orders in My Orders until the order ships. Once it has shipped, the tracking details are back in your spam folder."	Wrong: fluent but invented details
3	My package arrived broken, what should I do?	Report within 48 h with a photo; free replacement or refund	(empty)	"We are sorry about that! Report it within 48 hours of delivery with a photo through My Orders and we will send a free replacement or a refund."	Correct
4	Do you ship to Canada?	Yes, 40+ countries, 7–14 business days, customer pays duties	(empty)	"We deliver to over 50 countries. International delivery takes 3-5 business days."	Wrong: right topic, wrong numbers
5	I forgot my password, help!	Forgot Password link, reset email	(empty)	"Click Forgot Password on the login page and enter your email. We will send a reset link, so please check your spam folder if	Correct

#	Query	Expected (gist)	Base model	Fine-tuned model	Verdict
				you do not see it."	
6	Is there free delivery?	Free standard shipping over $50	(empty)	"Yes, standard orders can go to a PO box. Express orders require a street address."	Wrong: answer of another topic
7	Can I pay with PayPal?	Yes: Visa, MasterCard, Amex, PayPal, Apple Pay, gift card	(empty)	"We accept Visa, MasterCard, American Express, PayPal, Apple Pay and ShopEase gift cards."	Correct
8	How long will it take to get my money back after a return?	5–7 business days to the original payment method	(empty)	"Return shipping costs $4.99 and arrives within 3-5 business days."	Wrong: mixes two policies
9	What is the capital of France?	Out of scope: only orders, shipping, returns, accounts	(empty)	"We offer free size and colour exchanges within 30 days of delivery. Choose Exchange Item in My Orders to begin."	Wrong: no refusal
10	Can I change my delivery address after placing the order?	Only before it ships, in My Orders	(empty)	"You can edit your shipping address in My Orders before the order ships. Once it has shipped, the address can no longer be changed."	Correct

Result: 4 correct, 1 partly correct and 5 wrong answers out of 10 for the fine-tuned model. The base model gave no usable answer to any query (empty output; on the 81 test questions it occasionally produced a single word such as "Yes" or "Yes, we can.").

What fine-tuning changed. The fine-tuned model speaks like the ShopEase assistant: short policy-style sentences, the store vocabulary ("My Orders", "ShopEase gift cards", "Forgot Password") and a clean end of answer. Questions that closely match a training topic (damaged item, password reset, payment methods, address change) are answered correctly.

Failure modes.

- *Wrong-topic answers:* "Is there free delivery?" receives the PO-box answer, and the 2FA question receives the exchange answer. The first generated words decide the topic, and a weak question-to-answer association leads to a confident but unrelated answer.
- *Blended or invented facts:* "over 50 countries" and "3-5 business days" for international shipping, "Return shipping costs $4.99 and arrives within 3-5 business days" for refund timing, and "express shipping is available for orders above $50 ... free for orders above 50 points" on the test set. Pieces of different training answers are recombined into fluent but false statements.
- *No refusal outside the domain:* the capital-of-France question gets a return-policy answer, although out-of-scope refusals exist in the training data.
- *Sensitivity to wording:* the weakest test cases by embedding similarity include "Do you support 2FA? Thanks!", "Hey, are your products fake or original?" and "Can I change the items in my order?". Random test examples show the same behaviour, for example "Where is the size chart?" answered with the price-adjustment policy and "Do you have express shipping?" answered with an invented price rule.

4.3 Merged model

The adapter was also merged into the base weights to create a stand-alone model that loads with a plain `AutoModelForCausalLM`; on three test queries it reproduced the outputs of the adapter model exactly.

5. Inference script

`inference.py` loads the base model and the saved LoRA adapter and offers three modes: a single query (`--query`), an interactive command-line loop, and a Gradio web interface (`--gradio`). It was tested end to end in a separate process: for "Where can I find my invoice?" it returned the correct answer (the invoice is emailed with the order confirmation and can be downloaded from My Orders), while "Can I return a product after 30 days?" returned the item-condition answer already seen in Section 4.2. A Gradio demo with a temporary public link was also launched from the notebook.

6. Challenges and limitations

- **Synthetic, template-based data.** The questions are paraphrases of a fixed set of topics and the test questions come from the same topics, so the scores say how well the model handles new wordings of known questions, not how it would handle real customer traffic. Real data would be noisier, and would also contain questions with no matching FAQ.
- **Few examples per topic.** About a dozen training pairs per topic, a six-layer model and only 2.8 % trainable parameters are a small basis for learning a precise question-to-policy mapping. The model learned the style and the vocabulary quickly but is unreliable in choosing the right policy, as the wrong-topic and blended answers show.
- **Hallucinated details.** Numbers, durations and conditions can be wrong while the sentence still sounds authoritative. In a real deployment answers would have to be checked against the policy text.
- **Weakness of the base-model comparison.** The base model's generated text begins with a line break, and the notebook keeps only the first line, which leaves most base answers empty. This

inflates the gap in BLEU, ROUGE and cosine similarity. The perplexity and the loss comparison are unaffected. Stripping leading whitespace before taking the first line would give a fairer baseline and should be applied if the notebook is re-run.

- **Metric limitations.** BLEU and ROUGE reward word overlap and penalise equally correct answers phrased differently (each topic has two answer wordings), while embedding similarity cannot reliably detect a wrong number inside an otherwise similar answer. The manual evaluation is therefore the more informative result.
- **Small test set.** 81 test questions come from only about 33 base questions, so all figures are noisy and based on a single run with one seed.
- **Quantisation.** 8-bit and 4-bit loading was provided as an option but not used, because it does not apply to the GPT-2 layer type and the model is small enough for fp32 weights.

7. Conclusions and future work

LoRA fine-tuning of DistilGPT-2 on 671 synthetic ShopEase pairs took only 34 seconds on a T4 GPU and changed the model from a generic text generator into an assistant that writes concise, policy-style answers: the perplexity of the reference answers fell from 41.1 to 2.48 and all generation metrics improved (BLEU 14.7, ROUGE-L 0.27, embedding similarity 0.52). In the manual test it answered 4 of 10 queries correctly and 1 partly, and its main weaknesses are choosing the wrong policy, mixing facts from different answers and failing to refuse out-of-scope questions.

Promising next steps, which were not tested in this work: train longer or with a higher learning rate and rank, since the run took well under a minute; use a larger model within the limit such as `gpt2` (124 M) or `facebook/opt-125m` (which also allows 4-bit QLoRA); add more paraphrases per topic and more out-of-scope examples; and combine the model with retrieval of the matching FAQ entry so that the generated answer is grounded in the policy text. Evaluation could also include a fact-level check (does the answer contain the right numbers) and several random seeds.

Appendix – Files delivered with the report

Notebook `M5_LLM_FineTuning_FAQ_Assistant.ipynb`;

dataset `ecommerce_faqs.jsonl` and

generator `build_ecommerce_faq_dataset.py`;

cleaned splits `train.jsonl`, `val.jsonl`, `test.jsonl`;

checkpoints and the LoRA adapter (`lora_adapter/`);

`inference.py`; `automatic_metrics.csv`, `manual_evaluation.csv`, `test_predictions.csv`, `training_log.csv`; figures `fig_dataset_lengths.png`, `fig_training_curves.png`, `fig_metrics_base_vs_ft.png`; `README.md`.